

```

37 -- step 2
38 CREATE OR REPLACE TRIGGER sal_audit_2
39 BEFORE UPDATE OF sal ON emp
40 FOR EACH ROW
41 DECLARE
42     v_max_sal number;
43 BEGIN
44
45     -- get max salary wont work because of mutating table error!
46     select max(sal) into v_max_sal from emp;
47
48     IF :new.sal <= v_max_sal THEN
49         INSERT INTO emp_log (empno, log_text, log_date) VALUES
50         (:new.id, 'Gehalt Update Erfolgreich - altes Gehalt: ' || :old.sal || ', neues Gehalt: ' || :new.sal, sysdate);
51     ELSE
52         INSERT INTO emp_log (empno, log_text, log_date) VALUES
53         (:new.id, 'Gehalt nicht geändert - kann nicht mehr als ' || v_max_sal || ' sein.', sysdate);
54         :new.sal := :old.sal; -- Verhindert die Aktualisierung des Gehalts
55     END IF;
56 END;
57 /
58
59 UPDATE emp SET sal = 160000 WHERE id = 1;
60
61 Select * from emp_log; -- Überprüfen der Log-Einträge
62 select * from emp; -- Überprüfen der Gehälter in der emp Tabelle

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SCRIPT OUTPUT SQL HISTORY TASK MONITOR QUERY RESULTS SERIAL MONITOR

```

Error starting at line : 1 in command -
UPDATE emp SET sal = 160000 WHERE id = 1
Error at Command Line : 1 Column : 8
Error report -
SQL Error: ORA-04091: table ITMAM.EMP is mutating, trigger/function may not see it
ORA-06512: at "ITMAM.SAL_AUDIT_2", line 6
ORA-04088: error during execution of trigger 'ITMAM.SAL_AUDIT_2'

```

<https://docs.oracle.com/error-help/db/ora-04091/04091.00000-> "table %s.%s is mutating, trigger/function may not see it"

```

62 -- step 3
63
64 create or replace package emp_pkg is
65 | g_max_sal emp.sal%type;
66 end emp_pkg;
67 /
68
69 create or replace package body emp_pkg is end emp_pkg;
70 /
71
72 -- trigger on package variable to update the max salary
73 create or replace trigger sal_audit_stmt
74 before update of sal on emp
75 begin
76 | select max(sal)
77 | into emp_pkg.g_max_sal
78 | from emp;
79 end;
80 /
81
82 CREATE OR REPLACE TRIGGER sal_audit_3
83 BEFORE UPDATE OF sal ON emp
84 FOR EACH ROW
85 BEGIN
86 | -- use the package variable instead of querying the table
87 | IF :new.sal <= emp_pkg.g_max_sal THEN
88 | INSERT INTO emp_log (empno, log_text, log_date)
89 | VALUES (
90 | | :new.id,
91 | | 'Gehalt Update Erfolgreich - altes Gehalt: ' ||
92 | | :old.sal || ', neues Gehalt: ' || :new.sal,
93 | | sysdate
94 | | );
95 | ELSE
96 | INSERT INTO emp_log (empno, log_text, log_date)
97 | VALUES (
98 | | :new.id,
99 | | 'Gehalt nicht geändert - kann nicht mehr als ' ||
100 | | emp_pkg.g_max_sal || ' sein.',
101 | | sysdate
102 | | );
103 |
104 | :new.sal := :old.sal;
105 | END IF;
106 END;

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SCRIPT OUTPUT SQL HISTORY TASK MONITOR QUERY RESULTS SERIAL MONITOR

EMPNO LOG_TEXT

```

-----
1 Gehalt nicht geändert - kann nicht mehr als 150000 sein.

```

ID	NAME	JOB	SAL
1	Alex	CEO	150000
2	Klaus	Sr RnD	75000
3	Rudi	Jr RnD	55000
4	Berta	Manager	90000
5	Fritz	Dev	60000

```
1 DROP TABLE emp_log;
2 DROP TABLE emp;
3
4 CREATE TABLE emp (
5     id number not NULL,
6     name varchar2(50),
7     job varchar2(50),
8     sal number not null,
9     constraint emp_pk primary key (id)
10 );
11
12 INSERT INTO emp VALUES (1, 'Alex', 'CEO', 150000);
13 INSERT INTO emp VALUES (2, 'Klaus', 'Sr RnD', 75000);
14 INSERT INTO emp VALUES (3, 'Rudi', 'Jr RnD', 55000);
15 INSERT INTO emp VALUES (4, 'Berta', 'Manager', 90000);
16 INSERT INTO emp VALUES (5, 'Fritz', 'Dev', 60000);
17
18 create table emp_log (
19     empno number,
20     log_text varchar2(400),
21     log_date date
22 );
23
24 -- testing
25
26 DROP TRIGGER sal_audit_1;
27 DROP TRIGGER sal_audit_2;
28 DROP TRIGGER sal_audit_3_stat;
29 DROP TRIGGER sal_audit_3;
30 -- DROP TRIGGER sal_audit_4;
31
32 -- test step 3
33 UPDATE emp SET sal = 170000 WHERE id = 2;
34
35 select * from emp_log; -- Überprüfen der Log-Einträge
36 select * from emp; -- Überprüfen der Gehälter in der emp Tabelle
```

```
1 -- step 4 - compound trigger
2
3 CREATE OR REPLACE TRIGGER sal_audit_4
4 FOR UPDATE OF sal ON emp
5 COMPOUND TRIGGER
6
7 -- Variable, um höchsten SAL vor dem Update zu speichern
8 g_max_sal NUMBER;
9
10 BEFORE STATEMENT IS
11 BEGIN
12     SELECT MAX(sal)
13     INTO g_max_sal
14     FROM emp;
15 END BEFORE STATEMENT;
16
17 BEFORE EACH ROW IS
18 BEGIN
19     IF :new.sal <= g_max_sal THEN
20         INSERT INTO emp_log (empno, log_text, log_date)
21         VALUES (
22             :new.id,
23             'Gehalt Update Erfolgreich - altes Gehalt: ' || :old.sal || ', neues Geh
24             SYSDATE
25         );
26     ELSE
27         INSERT INTO emp_log (empno, log_text, log_date)
28         VALUES (
29             :new.id,
30             'Gehalt nicht geändert - kann nicht mehr als ' || g_max_sal || ' sein.',
31             SYSDATE
32         );
33     END IF;
34     :new.sal := :old.sal; -- verhindert Aktualisierung
35 END IF;
36 END BEFORE EACH ROW;
37
38 END sal_audit_4;
39
40 /
41 commit;
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SCRIPT OUTPUT SQL HISTORY TASK MONITOR QUERY RESULTS SERIAL MONITOR

EMPNO LOG_TEXT

2 Gehalt Update Erfolgreich - altes Gehalt: 75000, neues Gehalt: 160000
2 Gehalt nicht geändert - kann nicht mehr als 160000 sein.

ID	NAME	JOB	SAL
1	Alex	CEO	150000
2	Klaus	Sr RnD	160000
3	Rudi	Jr RnD	55000
4	Berta	Manager	90000